# S. M. OSAID RIZVI

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To contribute to a knowledge-driven organisation by applying my biotechnology background and training in emerging technologies to analyse scientific information, explore innovation trends, and support informed decision-making through interdisciplinary research.

## EDUCATION

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| **CHANDIGARH UNIVERSITY** | B.E. in Biotechnology | CGPA – 8.31 | 2026 | Mohali, Punjab, India |
| **GYAN NIKETAN** | 10+2, Medical (CBSE) | 89% | 2021 | Patna, Bihar, India |
| **RPS RESIDENTIAL SCHOOL** | $10^{th}$ (CBSE) | 92.4% | 2019 | Patna, Bihar, India |

## INTERNSHIPS

**ACADEMOR – NANOTECHNOLOGY INTERNSHIP (IIT BOMBAY TECHFEST)** **(Aug - Sep 2023)**

- Synthesised silver and zinc oxide nanoparticles via green synthesis, optimising yield efficiency by 18% over baseline.
- Designed experimental workflows for 2 nanoparticle projects, integrating statistical analysis for reproducibility.
- Characterised nanoparticle antimicrobial potential against bacterial strains, validating 30% higher inhibition zones.
- Proposed industrial application of synthesised nanoparticles in eco-friendly biomedical coatings, bridging lab-to-market concepts.
- Strengthened data visualisation skills by presenting results to faculty & IIT Bombay mentors, receiving positive evaluation.

**SOCIAL INTERNSHIP – COMMUNITY DEVELOPMENT | CHANDIGARH UNIVERSITY** **(Jun - Jul 2023)**

- Surveyed 200+ rural households, identifying gaps in sanitation, education, and livelihood.
- Proposed 3 infrastructural solutions (improved road access, sanitation hubs, vendor stalls) presented to community leaders.
- Designed awareness drives supporting Swachh Bharat Abhiyan, increasing hygiene compliance by 40% (survey follow-up).
- Supported 10+ small-town vendors with financial resource mapping, improving income sustainability.
- Delivered a final report with statistical data & graphs, now used as a case reference for university rural engagement projects.

## TRAININGS

**KHUSHRU MEDICARE – QA/QC LABORATORY INTERN | SOLAN** **(May - Jul 2025)**

- Spearheaded WHO & GMP compliance checks across 5 pharma production batches, ensuring 100% adherence to sterility standards.
- Conducted >40 sterility assays and packaging inspections, reducing contamination risk by 15% compared to baseline reports.
- Monitored quality assurance protocols across multiple dosage forms, directly supporting batch release decisions.
- Assisted in documenting deviations and corrective actions, streamlining audit readiness for upcoming WHO inspections.
- Enhanced operational efficiency by standardising packaging inspection checklists, minimising manual errors during QC.

**TATA MAIN HOSPITAL – PATHOLOGY (MICROBIOLOGY & BIOCHEMISTRY) INTERN (May - Jul 2024)**

- Performed >100 biochemical assays (LFTs, Lipid Profiles, CRP, HbA1c, Vitamin B12) using Beckman Coulter DxC 700 AU, ensuring accurate diagnostics.
- Utilised BACT/ALERT 3D & VITEK 2 analysers for 50+ blood cultures, accelerating pathogen ID and antibiotic sensitivity by 25%.
- Conducted ELISA & rapid diagnostic tests for dengue, HIV, and HCV, contributing to early detection in 200+ patient samples.
- Collaborated with microbiologists to identify >12 bacterial strains in pus cell cultures, strengthening clinical reporting.
- Improved sample workflow by assisting in the collection-to-report cycle reduction from 72 hrs to 60 hrs.

## PROJECTS

**ECO-FRIENDLY BIODEGRADABLE GLOVES (Jan - Jun 2025)**

- Designed prototype gloves from biodegradable biopolymers, targeting the reduction of plastic waste in laboratories.
- Incorporated antimicrobial nanocoating, extending glove safety lifespan by 15% compared to regular disposables.
- Conducted tensile strength testing on 10+ samples, confirming comparable durability to nitrile gloves.
- Presented solution at departmental pitch competition, securing 2nd place out of 25 teams.
- Drafted patent feasibility analysis, highlighting potential scalability for academic & medical labs.

**HYDROXYAPATITE NANOPARTICLES FOR ALZHEIMER'S DRUG DELIVERY (Aug - Dec 2024)**

- Reviewed and analysed 30+ scientific studies on HAp nanoparticles for crossing the blood-brain barrier (BBB).
- Highlighted drug loading/release kinetics, noting 20–30% improvement in sustained release vs conventional carriers.
- Proposed functionalization strategies (antibody & peptide conjugation) to enhance BBB penetration.
- Evaluated hybrid nanoparticle approaches integrating HAp with immunotherapy, showing potential for 30–40% plaque reduction in models.
- Contributed to award-winning thesis presentation (Best Paper, ICNMGT-2024), later cited in institutional repository.

## ACHIEVEMENTS

- Best Thesis Paper Presentation Award – ICNMGT-2024
- 2nd Place – Oral Abstract Presentation at ICCMAS-2024
- 2nd Place – Start-Up Pitch Competition (Biodegradable Gloves)
- 2nd Prize – District Poetry Contest

## SKILLS

**TECHNICAL:**

- Data Analysis & Interpretation
- Laboratory Testing & Diagnostics
- Quality Assurance & Compliance (GMP/WHO)
- Sterility Testing
- Clinical Biochemistry
- Microbiological Analysis
- Bacterial Culturing
- Nanoparticle Synthesis

**TOOLS:**

- Power BI
- Microsoft Office Suite
- ELISA Analyser
- VITEK Analyser
- BACT/ALERT 3D
- Beckman Coulter DxC 700 AU

**PROFESSIONAL:**

- Project Management
- Research Presentation
- Strategic Leadership
- Team Collaboration & Coordination
- Remote Work Adaptability

**INTERPERSONAL:**

- Stakeholder Management
- Cross-functional Collaboration
- Critical Thinking & Problem-Solving
- Empathy & Resilience
- Adaptability

## CERTIFICATIONS

- Foundations of Project Management
- National Intellectual Property (IP) Awareness Workshop
- Industrial Biotechnology
- Introduction to Artificial Intelligence (AI)
- Genes and the Human Condition

## INTERESTS & HOBBIES

Writing | Sports | Painting | Travelling | Reading